DR. MANISHA SINGH

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Cornell Executive MBA Detailed CV | Science, AI, Wellness & Entrepreneurial Leadership

PROFESSIONAL PROFILE

PhD-trained biomedical scientist and cancer researcher with more than 15 years of experience in cell biology, immunology, oncology research, molecular biology, imaging-oriented translational research, teaching, mentoring, and collaborative laboratory environments. Currently translating scientific training into a next-stage leadership path focused on AI-supported research services, science-aware wellness education, and responsible entrepreneurship. Brings rigorous scientific discipline, global adaptability, resilience, and a distinctive perspective for an experienced MBA cohort.

SELECTED APPLICANT HIGHLIGHTS

- PhD-trained scientist with deep biomedical research experience across cancer biology, immunology, molecular biology, cell biology, and translational oncology.
- Author/co-author of 9 peer-reviewed publications and 1 supplementary research material/data contribution, with work spanning cancer immunotherapy, molecular imaging, dystroglycan biology, and protein trafficking.
- Collaborative researcher with experience across MSKCC, Albert Einstein College of Medicine, Stanford University, California Pacific Medical Research Institute, Erasmus Medical Center, and San Jose State University.
- Global, cross-cultural background across India, Europe, and the United States, including both East Coast and West Coast academic and research environments.
- Current professional direction: build science-aware wellness education and AI-supported research services through Shyaama Wellness Intelligence while preparing for business leadership and execution.

MBA FIT AND CAREER DIRECTION

- Purpose for the MBA: convert advanced scientific training into stronger leadership, strategy, finance, entrepreneurship, team-management, and execution capacity.
- Career direction: build and lead ventures or programs at the intersection of science, wellness education, AI-supported research, and responsible health innovation.
- Weekend MBA fit: the format is strategically aligned with current goals because weekdays can be used for preparation, learning-team collaboration, networking, and immediate application to Shyaama Wellness Intelligence.
- Cohort contribution: brings PhD-level scientific discipline, translational research experience, global perspective, resilience, AI learning, and founder-level initiative to team-based learning.
- Long-term leadership theme: move beyond a narrowly laboratory-based role toward broader organizational impact in science, wellness education, AI-enabled knowledge work, and business execution.

EDUCATION

PhD, Cell Biology and Immunology | Erasmus Medical Center | Rotterdam, The Netherlands | 2013

- Advisors: Frank Grosveld, PhD; Claudia Gravekamp, PhD.
- Research training included cell biology, immunology, tumor immunology, disease mechanisms, and translational biomedical investigation.

Master of Science, Molecular Biology and Microbiology | California State University, San Jose | San Jose, CA | 2006

- Advisor: David J. Matthes, PhD.
- Graduate training included molecular biology, microbiology, genetics, cell biology, and laboratory teaching responsibilities.

Master of Science, Biological Sciences | Purvanchal University | India | 1999

Bachelor of Science, Biology and Chemistry | Purvanchal University | India | 1997

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Fellow, Department of Radiology | Memorial Sloan Kettering Cancer Center | New York, NY | 2014-2016

- Conducted oncology-focused translational research involving molecular imaging, radiopharmaceutical approaches, prostate cancer biology, and cancer-relevant signaling mechanisms.

- Developed a novel PET tracer for imaging androgen receptor-independent lesions in metastatic prostate cancer.
- Characterized radiation-induced nuclear translocation of the EGF/EGFR complex and contributed to mechanistic understanding of cancer-related signaling responses.
- Worked with PET imaging approaches using I-124 and Zr-89 radiolabeled antibodies in xenografted mouse models to target proteins commonly overexpressed in cancer.
- Prepared and presented research findings for scientific audiences, including Annual World Molecular Imaging Congress presentations.
- Contributed to manuscript preparation, collaborative data interpretation, scientific communication, and translational research planning.

PhD Thesis Research, Department of Microbiology and Immunology | Albert Einstein College of Medicine | Bronx, NY | 2009-2013

- Investigated cancer immunotherapy strategies in metastatic breast cancer models, with emphasis on NKT-cell activation, Listeria-based vaccine platforms, curcumin, and T-cell responses.
- Studied how alpha-galactosylceramide activation of NKT cells could improve Listeria-Mage-b vaccine-based immunotherapy against metastatic breast cancer.
- Evaluated the effect of curcumin in combination with Listeria-Mage-b vaccination in triple-negative breast cancer models and correlated therapeutic response with immune-cell activity.
- Designed and executed immunological analyses using blood, spleen, lymph node, tumor, and metastasis samples for FACS and related assays.
- Generated peer-reviewed publications and conference presentations in cancer immunotherapy and metastatic breast cancer research.

PhD Student, Department of Cell Biology | Erasmus Medical Center | Rotterdam, The Netherlands | 2009-2010

- Investigated the role of Filamin A in microtubule dynamics and its relationship to periventricular nodular heterotopia, integrating cell biology with human disease mechanisms.
- Developed cross-institutional research adaptability through international doctoral training and biomedical research collaboration.

Research Assistant | California Pacific Medical Research Institute | San Francisco, CA | 2006-2009

- Studied dystroglycan processing and function in normal and cancerous cells, including structural modeling, mutagenesis, protein trafficking, and mammary epithelial biology.
- Discovered and characterized nuclear trafficking of beta-dystroglycan using GFP-fusion protein tracking and cell fractionation approaches.
- Investigated endocytosis and turnover of basement membrane proteins using flow cytometry, histology, immunohistochemistry, and tissue analyses from knockout mouse models.
- Contributed to multiple peer-reviewed publications in cell biology, cancer-related signaling, dystroglycan biology, and protein trafficking mechanisms.
- Trained and mentored new laboratory members in molecular and cell biology techniques.

Research Assistant / Visiting Scholar, Department of Developmental Biology | Stanford University | Stanford, CA | 2005-2006

- Conducted molecular genetics and bioinformatics analyses of insulin orthologs across metazoans, invertebrate models, and mammalian systems.
- Built early cross-disciplinary experience linking developmental biology, genetics, comparative biology, and computational analysis.

TEACHING, MENTORING, AND TRAINING EXPERIENCE

Teaching Assistant, Department of Biological Sciences | San Jose State University | San Jose, CA | 2005-2006

- Designed and taught laboratory classes in Human Biology and Cell Biology, supporting student learning in experimental techniques and biological concepts.
- Developed instructional experience relevant to adult learning, scientific communication, classroom leadership, and mentoring.

Graduate Assistant / Graduate Student Researcher | San Jose State University | San Jose, CA | 2003-2005

- Mentored undergraduate students conducting genetic analysis of Drosophila sema1b.
- Taught laboratory students to produce a monoclonal antibody against Listeria monocytogenes.
- Supported research training, laboratory organization, and student development in biological sciences.

TEAMWORK, COLLABORATION, AND LEADERSHIP EXPERIENCE

- Developed strong teamwork skills through biomedical research environments where progress depended on collaboration among principal investigators, postdoctoral researchers, graduate students, research assistants, technicians, and external collaborators.

- Collaborated closely with postdoctoral researchers and laboratory teams during research assistant, PhD, and postdoctoral work, contributing to experimental planning, laboratory execution, data organization, troubleshooting, and shared project goals.
- Coordinated cross-departmental research activities with shared-resource and specialized teams, including imaging, tissue preparation/histology-related support, animal-model support, and technical core facilities.
- Regularly worked across more than one project at a time, including projects within the primary laboratory and projects with collaborating laboratories or research groups.
- Mentored, trained, and supported junior laboratory members and students in molecular biology, cell biology, experimental techniques, troubleshooting, documentation, and interpretation of research results.
- Comfortable working in cohort-based and project-based teams; brings disciplined preparation, careful listening, clear communication, respect for diverse expertise, and commitment to shared outcomes.

GLOBAL PERSPECTIVE AND CROSS-CULTURAL ADAPTABILITY

- Lived, studied, and worked across three continents, gaining direct exposure to Indian, European, and American academic, scientific, and cultural environments.
- Developed cross-cultural adaptability through experience navigating different education systems, research cultures, communication styles, and professional expectations.
- Within the United States, lived and worked on both the East Coast and West Coast, gaining familiarity with regional differences in academic, research, and professional environments.
- Traveled across India, Europe, and the United States, strengthening cultural awareness, flexibility, and ability to work with diverse teams.

INDEPENDENT STUDY, PROFESSIONAL REBUILDING, AND CAREER TRANSITION

- Engaged in a period of professional rebuilding and independent learning following an interruption in the biomedical research career path.
- Continued study in whole-person wellness, health systems, Ayurveda-informed wellness concepts, nutrition/lifestyle thinking, and broader healing frameworks while maintaining a science-aware and evidence-conscious perspective.
- Explored the limitations of highly specialized health-care silos and developed a long-term interest in whole-person wellness models that consider the body, mind, lifestyle, environment, and prevention-oriented education together.
- Built skills in AI-supported research organization, prompt-based knowledge work, document development, strategic planning, and digital business systems.
- Clarified a future leadership goal: bridging scientific training, wellness education, AI tools, and business execution.

ENTREPRENEURIAL AND AI-ENABLED BUSINESS DEVELOPMENT

Founder-Initiated Venture / Pre-Launch Development | Shyaama Wellness Intelligence | New York, NY / Remote | 2026-Present

- Developing Shyaama as a broad lifestyle and wellness knowledge platform, beginning with science-aware wellness education and AI-supported research services rather than inventory-heavy operations.
- Created initial service positioning for Shyaama Science and AI Research Support, including literature review support, scientific evidence summaries, wellness claim review for education, AI-assisted research workflows, and science-aware content support.
- Built an internal AI-assisted operating system concept for founder planning, documentation, governance preparation, time tracking, service development, and launch readiness.
- Prepared launch materials for a soft website launch, including service descriptions, founder bio, pilot package language, intake questions, and educational disclaimers.
- Maintains careful distinction between education/research support and medical advice, diagnosis, or treatment.
- Coordinates strategic documentation, business planning, brand development, and team communication while preparing for professional accounting/legal review as needed.

SELECTED LEADERSHIP, TEAMWORK, AND TRANSFERABLE SKILLS

Skill Area	Evidence / Application
Strategic thinking	Connects laboratory research, health/wellness needs, AI tools, and emerging business models.
Research discipline	Defines hypotheses, designs experiments, interprets complex data, troubleshoots uncertainty, and communicates evidence responsibly.
Communication	Experienced in scientific presentations, manuscript preparation, meeting documentation, email communication, and structured explanations for non-specialist audiences.
Mentoring and teaching	Taught undergraduate labs, trained students and laboratory members, and translated complex biological concepts into teachable steps.
Cross-cultural perspective	Educational and professional experience across India, the Netherlands, and the United States.
MBA learning-team readiness	Prepared to contribute consistently to weekend classes, weekday team meetings, collaborative assignments, peer learning, and cohort-based problem solving.

SCIENTIFIC AND TECHNICAL SKILLS

Category	Skills / Experience
Biomedical Research	Oncology research; cancer biology; tumor immunology; cell biology; molecular biology; immunology; translational research; radiopharmaceutical and PET imaging research.
Experimental Design and Analysis	Hypothesis development; experimental planning; biological data interpretation; troubleshooting; research methodology; manuscript preparation.
Molecular Biology	Recombinant DNA/RNA techniques; purification; PCR; cloning; mutagenesis; cDNA library construction and screening.
Cell Biology and Cell Culture	Primary and established cell culture; mammary epithelial cells; transfection; fluorescence-activated cell sorting/FACS; protein trafficking assays.
Biochemistry	Protein purification; affinity chromatography; immunoaffinity chromatography; SDS-PAGE; Western blotting; protein-protein interaction assays.
Histology and Imaging	Cryosectioning; immunofluorescence; immunohistochemistry; histochemical staining; confocal microscopy; epifluorescence microscopy.
Immunological Techniques	ELISPOT; extracellular and intracellular staining; blood, spleen, lymph node, tumor, and metastasis sample preparation for FACS analysis.
Animal Models	Mouse handling; immunization; organ dissection; single-cell suspension preparation from spleen, lymph nodes, tumors, metastases, and blood.
Computational / Digital	Bioinformatics database searching; BLAST; multiple sequence alignment; primer design; plasmid annotation; Microsoft Word, Excel, PowerPoint; AI-assisted research organization and document workflows.

PUBLICATIONS AND RESEARCH OUTPUTS

Peer-Reviewed Publications

- Cheal SM, Xu H, Guo HF, Patel M, Punzalan B, Fung EK, Lee SG, Bell M, Singh M, Jungbluth AA, Zanzonico PB, Piersigilli A, Larson SM, Cheung NKV. Theranostic pretargeted radioimmunotherapy of internalizing solid tumor antigens in human tumor xenografts in mice: Curative treatment of HER2-positive breast carcinoma. Theranostics. 2018. doi: 10.7150/thno.26585. PMID: 30429889; PMCID: PMC6217068.
- Jahangir A, Chandra D, Quispe-Tintaya W, Singh M, Selvanesan BC, Gravekamp C. Immunotherapy with Listeria reduces metastatic breast cancer in young and old mice through different mechanisms. Oncoimmunology. 2017 Jul 5;6(9):e1342025. doi: 10.1080/2162402X.2017.1342025. PMID: 28932647.
- Leonoudakis D, Huang G, Akhavan A, Fata JE, Singh M, Gray JW, Muschler JL. Endocytic trafficking of laminin is controlled by dystroglycan and is disrupted in cancers. J Cell Sci. 2014 Nov 15;127(Pt 22):4894-903. doi: 10.1242/jcs.152728. PMID: 25217627.

- Singh M, Quispe-Tintaya W, Chandra D, Jahangir A, Venkataswamy MM, Ng TW, Sharma-Kharkwal S, Carreno LJ, Porcelli SA, Gravekamp C. Direct incorporation of the NKT-cell activator alpha-galactosylceramide into a recombinant Listeria monocytogenes improves breast cancer vaccine efficacy. Br J Cancer. 2014 Nov 11;111(10):1945-54. doi: 10.1038/bjc.2014.486. PMID: 25314062.
- Singh M, Ramos I, Asafu-Adjei D, Quispe-Tintaya W, Chandra D, Jahangir A, Zang X, Aggarwal BB, Gravekamp C. Curcumin improves the therapeutic efficacy of Listeria(at)-Mage-b vaccine in correlation with improved T-cell responses in blood of a triple-negative breast cancer model 4T1. Cancer Med. 2013 Aug;2(4):571-82. doi: 10.1002/cam4.94. PMID: 24156030.
- Gravekamp C, Chandra D, Quispe W, Jahangir A, Singh M, et al. New approaches in cancer vaccination at old age. Journal of Geriatric Oncology. 2012 Oct.
- Leonoudakis D, Singh M, Mohajer R, Mohajer P, Fata JE, Campbell KP, Muschler JL. Dystroglycan controls signaling of multiple hormones through modulation of STAT5 activity. J Cell Sci. 2010 Nov 1;123(Pt 21):3683-92. doi: 10.1242/jcs.070680. PMID: 20940259.
- Oppizzi ML, Akhavan A, Singh M, Fata JE, Muschler JL. Nuclear translocation of beta-dystroglycan reveals a distinctive trafficking pattern of autoproteolyzed mucins. Traffic. 2008 Dec;9(12):2063-72. doi: 10.1111/j.1600-0854.2008.00822.x. PMID: 18764929.
- Akhavan A, Crivelli SN, Singh M, Lingappa VR, Muschler JL. SEA domain proteolysis determines the functional composition of dystroglycan. FASEB J. 2008 Feb;22(2):612-21. doi: 10.1096/fj.07-8354com. PMID: 17905726.

Supplementary Research Material / Data Contribution

- Supplementary research material/data contribution associated with Cheal et al., Theranostics 2018, October 2018. Listed separately from peer-reviewed publications.

Manuscripts and Selected Presentations

- Singh M, Chakroborty G, Medine EI, Kalidindi TM, Lee SG, Punzalan B, Giancotti F, Pillarsetty NVK, Larson SM. Novel PET tracer for imaging androgen receptor-independent lesions in metastatic prostate cancer. Manuscript in preparation.
- Singh M, Medine EI, Ku A, Kalidindi TM, Lee SG, Punzalan B, Pillarsetty NVK, Larson SM. Radiolabeled EGF derivatives as radiotherapy enhancers. Manuscript in preparation.
- Singh M, Chakroborty G, Medine EI, Kalidindi TM, Lee SG, Punzalan B, Giancotti F, Larson SM, Pillarsetty NVK. Progress towards the development of a novel immunoPET tracer for imaging androgen deprivation therapy lesions in metastatic prostate cancer. Annual World Molecular Imaging Congress, New York, NY, September 2016.
- Singh M, Kalidindi TM, Ku A, Lee SG, Punzalan B, Boyko V, Turkekul M, Larson SM, Pillarsetty NVK. Characterization of radiation-induced nuclear translocation of EGF/EGFR complex. Annual World Molecular Imaging Congress, Honolulu, HI, September 2015.
- Singh M, Ramos I, Adjei D, Quispe W, Chandra D, Jahangir A, Aggarwal B, Gravekamp C. Improvement of efficacy by Listeria-Mage-b vaccination through curcumin. New York Cancer Consortium Developmental Therapeutics Symposium, New York, NY, October 2011.
- Singh M, Ramos I, Adjei D, Quispe W, Chandra D, Jahangir A, Aggarwal B, Gravekamp C. Vaccination with Listeria-Mage-b and curcumin is highly effective against metastatic breast cancer. 19th Annual CRI Cancer Immunotherapy Symposium, New York, NY, October 2011.
- Singh M, Grosveld F. The role of Filamin A in congenital brain disease. MGC Graduate Student Workshop, Cologne, Germany, June 2010.

PROFESSIONAL MEMBERSHIPS AND SERVICE

- Member, American Society for Microbiology.
- Member, American Society for Cell Biology.
- Member, New York Academy of Sciences.
- Purvanchal University Graduate Student Association representative, 1998.
- Volunteer, 16th Annual CSU Biotechnology Symposium, January 2004.

LANGUAGES

- English: professional scientific and business communication.
- Hindi: native/fluent communication.

REFERENCES

Available upon request.